# A 125B mixture-of-experts model in 32, 24 and 12 GB of memory on one Mac

SSD-streamed experts, a verified drafting loop at 12 GB, and where small memory loses: whole decode, whole request and first token for each memory line.

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**Abstract**

We run Qwen3.8-Flash-Next, a 125B mixture-of-experts model (48 layers, 512 routed experts per layer, 10 per token), on one MacBook Pro (M5 Pro, 18-core CPU, 20-core GPU, 64 GB, 2 TB SSD). Experts are streamed from the internal SSD. On our speed sets the process's own lifetime peak footprint stays under 32, 24 or 12 GB, except for one 520-token math prompt at 24 GB (24.01 GB); longer prompts go over the 24 and 32 GB lines (section 6.3).

**At 12 GB** a block drafter's 15 drafts are checked by the 12 GB model in calls of 2-4 rows. Experts are stored at about 3.2 bits per weight.

- The loop is **verified**: every committed token is the model's greedy argmax at its computed width. It is not bit-identical to one-token decoding.
- It is deterministic and independent of the cache contents and of the prompt order.
- On 12 HumanEval + 12 MBPP it runs 18.3 tok/s whole decode, 13.1 tok/s whole request and 4.3 s first token, at an 11.14 GB peak. On chat (12 IFEval) it runs 12.8 / 10.1 tok/s and 2.9 s, at 10.96 GB [K].

**At 24 GB** a lossy configuration ([MAIN]'s L + COH with 141 + 26 expert slots per layer and checks of at most 13 rows) runs 130.3 tok/s whole decode, 40.8 tok/s whole request and 3.54 s first token on 12 HumanEval, at a 23.82 GB peak [K].

- It passed its quality gate: the same scores as the exact reference on IFBench, GPQA Diamond and LiveCodeBench v6, in both orders [K].
- In our runs it stayed under 24 GB on prompts of up to about 500 tokens, with little room: a 448-token prompt reached 23.954 GB and a 520-token one 24.01 GB. Longer prompts use more: about 25.0 GB at 3K tokens, 25.8 GB at 10K and 28.2 GB at 30K [K].
- On 12 MBPP it runs 91.2 / 26.1 tok/s and 3.3 s, on chat (12 IFEval) 30.6 / 18.1 tok/s and 3.1 s, and on 12 math prompts 40.8 / 27.4 tok/s and 2.9 s [K].
- An older exact configuration (exact to its own stock-kernel reference) runs 41.1 tok/s whole decode on 12 HumanEval and 37.4 on 12 MBPP, at a 23.86 GB peak [K]. Its whole request and first token were not recorded.

**At 32 GB**, on chat (12 IFEval) [K]:

- The exact configuration, token-identical to the 45 GB reference on 12 / 12 prompts, runs 33.8 tok/s whole decode, 19.8 whole request and 2.8 s first token, at a 30.62 GB peak.
- A lossy chat configuration, not gated at 32 GB, runs 51.3 / 22.5 / 3.0 s at 31.22 GB.

**Where we lose.**

- Strata (MIT; github.com/Niko1221/Strata), an open-source engine for this model, cannot run at 12 GB: its smallest setting peaks at 14.19 GB.
- But at that setting its chat beats ours: 14.4 vs 12.8 tok/s decode, 11.2 vs 10.1 whole request, 2.8 vs 2.9 s first token.
- At 24 GB, on the same prompts, its first token (1.3-2.3 s) is ahead of ours (2.9-3.5 s). On chat its whole request is also ahead (20.3-21.0 vs 18.1 tok/s), and our decode leads only 1.1x. On code and math our whole decode (40.8-130.3 vs 26.1-28.0 tok/s) and whole request (26.1-40.8 vs 20.3-23.0) lead.
- Our 24 GB configuration can go over 24 GB on prompts longer than about 500 tokens: 25.0 GB at 3K tokens, 28.2 GB at 30K. On six long gate prompts (726-2,801 tokens), 256-token prompt-pass chunks kept it under, but cost 2.0-2.9x the first token and changed the tokens, so its gate does not cover them.
- Our 12 GB first token is 2.9-4.3 s, because the prompt pass streams experts from the SSD: about 22 GB for a 63-token prompt.

- Our 12 GB lossy two-pass mode failed its HumanEval+ gate at every threshold tried.

Two early headlines were wrong. "12 GB, 40.8 tok/s exact" was a lossy arm, and "the likely first at <= 12 GB, exact" falls with it.

We measure where a 12 GB verify call spends its time. Levers priced and not yet run:

- two-row verify calls on chat [P: +10-15 %]
- Strata's next-layer read-ahead
- a lossy 24 GB chat route that reads 81.7 MB of SSD per kept token instead of 210.2 [P]

# 1. Introduction

## 1.1 The setting

- **The model** is Qwen3.8-Flash-Next: 125B parameters, 48 layers, 512 routed experts per layer, 10 per token.
- **The store** has about 3.2 bits per expert weight: ranks 0-1 of a token's experts at 4 bits, ranks 2-9 at 3 bits (the top 3 bits of the 4-bit store).
- **The machine** is one MacBook Pro, M5 Pro, 64 GB unified memory, internal SSD. The sequential reads measured inside the 12 GB verify calls reach about 12 GB/s [K, F12P: 81.4 ms per GB].
- **A memory line of N GB** means that the process's lifetime peak footprint is at most N GB.
  - GB = `10^9` bytes of the lifetime peak (ri_lifetime_max_phys_footprint). The 12 GB line's 11.140 GB is 10.375 GiB [K, F12 'TABLE 12 GB CELLS 12E + 12F'].
  - This is the engine's own footprint on a 64 GB machine, not a limit the operating system enforces.
  - Reads use F_NOCACHE, so the page cache does not hold experts for us.
- **Why these lines:** 12, 24 and 32 GB are common memory sizes [G]. Companion paper [MAIN] covers 45 GB.

## 1.2 Contributions

1. A 12 GB configuration of a 125B MoE that is verified, deterministic and history-independent. It is measured end to end on code and chat, with the reasons it is not exact (section 2).
2. A 24 GB configuration that passed its quality gate (same scores on IFBench, GPQA Diamond and LiveCodeBench v6, both orders) and runs 130.3 tok/s whole decode on 12 HumanEval and 91.2 on 12 MBPP under 24 GB, with the prompt length up to which it stays under (sections 2, 3.4, 6.1 and 6.3).
3. A per-stage cost model of one 12 GB verify call, measured: reads, loader-idle time, main-thread CPU and the GPU side (section 7). It prices the levers left (section 11).
4. Same-machine comparisons with Strata at 24 GB and at its smallest setting, from our own runs (section 6.5).
5. The negative results at small memory: prefetch, budgets, read layouts, re-quantization, narrower drafts, and a lossy mode that passed our internal screen but failed the benchmark (section 8).
6. The decision map used to screen lossy rules (section 4) and the SSD-bytes model BYTES10 (section 5).

## 1.3 What we do not claim

- No "first" or "fastest". The 4 Nov "likely first at <= 12 GB, exact" claim is withdrawn (section 8).
- The 12 GB loop is not exact to one-token decoding, to the 4-bit weights, or to the 45 GB reference.
- The 12 GB lossy modes are not certified: the two-pass mode failed its gate, and the width rule passed only an IFBench test, on loose scoring; its strict score is significantly lower (section 2).
- The 24 GB configuration is lossy, and its gate covers IFBench, GPQA Diamond and LiveCodeBench v6. Its speed is measured on HumanEval, MBPP, chat (IFEval) and math; HumanEval flatters (section 6.1).
- Strata runs other weight files (3-bit IQ3_XXS) and is lossy in the arms that fit small memory. The comparison is same machine and same prompts, not the same weights or the same exactness class.

# 2. What each line computes

- **12 GB, verified** ([MAIN] section 2).
  - The 12 GB model checks the drafts in calls of 2-4 rows. Every routed rank is read and waited for, and every committed token is the model's greedy argmax at its computed width.
  - It is not exact. A width-1 forward differs from widths 2-4 at rounding level. The line's own one-token path defers cold ranks (section 3.2), so its bits depend on the cache [K, F12 'What the 12 GB exact loop (arm ex) is exact to'].

  - Independent of history [K]. In another prompt order, 13 / 13 prompts were bit-identical in tokens and in the hash of every kept row's logits. Each prompt's SSD bytes moved by -1.1 % to +12.5 % between the orders, so the cache contents really differed (F12H).
  - The benchmark gate saw the same thing: "the engine shows no history effect" (EV:RESULT_t80ck1u). One prompt order is therefore enough to gate this line.
- **The reference.** REF2-T is the exact reference the 12 GB gates use; it runs the line's tiers (EV:RESULT_t80ck1u). In the gate's VOID check, the verified loop was token-identical to REF2-T on 4 prompts x 64 tokens [K, same file].
- **12 GB two-pass FAST: lossy, failed its gate** [K, EV:RESULT_t80ck1u]. A cheap top-2 pass certifies rows; uncertain rows go to the full check. The test is HumanEval+ against REF2-T, 60 prompts, both orders:
  - theta 0.8: -11.67 pt (p 0.0156)
  - theta 0.9: -8.33 pt
  - theta 0.95: -5.00 pt, exactly at the bar, which needs a delta above -5. Against NAX6-N it is -1.67 pt.
- **12 GB confidence-adaptive first-call width (P2'): lossy at rounding level, same scores on IFBench on loose scoring (one order); strict scoring lower** [K, EV:RESULT_card_ifb12].
  - Every committed token is still the model's argmax, but the committed state depends on the call widths: tokens equal the 4-row reference on 5 / 12 chat prompts [K, F12 'T2 CHAT WIDTH stage 2'].
  - Its IFBench test against NAX6-N (150 prompts, one order, and a 10-prompt reverse check that matched; not a full card gate) passed on the gate's metric, loose instruction following: 40.0 % vs 44.0 %, -4.00 pt, p 0.21, against a line of a delta above -5 pt with p at least 0.05 [K, EV:RESULT_card_ifb12].
  - On strict instruction following it is significantly lower: 32.7 % vs 40.0 %, -7.33 pt, p 0.013. The 24 and 45 GB lossy configurations score 36.7 % strict on the same prompts. No claim of equal instruction following rests on this test [K, same file].
  - 16 of its 150 answers hit the 1,536-token cap, against 10 for the reference. 5 of them end in a near-pure loop: at least 87 % of the 8-token sequences in their last 512 tokens repeat [K, EV:RESULT_card_ifb12; F12 'IFB12 RUN'].
- **24 GB (w13b): lossy, same scores on IFBench, GPQA Diamond and LiveCodeBench v6 (both orders)** [K, EV:RESULT_card_w24].
  - It is [MAIN]'s lossy configuration L + COH, with smaller pools and checks of at most 13 rows (section 3.4). As at 45 GB, drafted rows on code prompts skip cold experts that would miss [K, BM:w24env].
  - Its gate against NAX6-N, the exact reference ([MAIN] section 2): IFBench 40.7 % vs 44.0 % and GPQA Diamond 47.0 % vs 44.0 % in both orders; LiveCodeBench v6 29.8 % and 32.8 % vs 32.1 % [K, EV:RESULT_card_w24].
  - On IFBench and GPQA Diamond its tokens equal those of the 45 GB L + COH gate on 250 / 250 prompts per order [K, same file].
- **24 GB (E24), exact to its own stock-kernel reference** [K, BM:WED 'E24'].
  - Two passes are identical on 24 / 24 prompts, and the run is exact against that reference.
  - Not exact to NAX6-N: 6 / 24 prompts agree by sha (information).
- **32 GB.**
  - e32 is the exact configuration E (exact to NAX6-N; [MAIN] section 2) at cache C 160. In this run its tokens equal NAX6-N's one-token decoding on 12 / 12 chat prompts; the run did not hash logits [K, BM:c32, BM:WED 'C32'].
  - zs32 is the chat configuration M1-B + COH + SB1 at C 160: lossy, not gated at 32 GB.
    * Its IFEval gate ran at C 256, and at C 160 it skips more cold ranks. It is token-identical on 2 / 12 prompts, as at C 256 [K, BM:WED 'C32'].
    * A 32 GB gate for it was queued last in the release day's machine chain, and did not fit before the 16:00 cutoff.

## 3. The system at small memory

Our engine builds on the model code of MTPLX by Youssof Altoukhi (Apache-2.0; github.com/youssofal/mtplx), installed from a pinned commit, for this architecture. MLX is pinned to 0.32.2 with the pad-to-64 fix for MLX issue #4632. [RELEASE] gives the package.

## 3.0 How the 12 GB route was found

The 12 GB line rests on three choices: which experts to give up, how many bits to keep, and how to decode. Below are the measurements behind each choice, including the two that turned out wrong.

**Dropping experts cost more than dropping bits (late October).**

- The first small-memory rule ("frugal") kept a tenth of the experts resident and dropped the cold routed ones. It failed 5 of the benchmark harness's 8 teacher-forced fidelity rules: KLD 5.97x the 4-bit reference's, top-1 agreement 3.50 points lower [K, SB '... verdicts on frugal']. Its decode tests stayed inside that day's bars (IFEval 88.0 vs 90.0 %, HumanEval 90.0 vs 91.7 %) [K, same section].
- A teacher-forced screen on the real 12 GB decoder then split the loss. Hard text = IFEval + multilingual, 7 sequences, 2,353 positions [K, SB 'QUALITY STEP screen on the real v3f path']:
  - the 12 GB store (ranks 2-9 at 3 bits) with cold ranks dropped: KLD 0.1117;
  - the same with the pools at 4 bits (1.1 GB more memory): 0.0993, -11 %;
  - the cold ranks 2-9 computed half a layer late instead of dropped, at the same memory: 0.0178, -84 %. That passes the strict per-part targets (IFEval 0.0186 vs 0.0204, multilingual 0.0170 vs 0.0206).
  - The record's conclusion: "The dominant quality cost of frugal is DROPPING cold ranks 2-9, not the 3-bit tiers".

**So the store keeps the tiers and never drops a routed rank.**

- Ranks 0-1 of each token's ten experts are served at 4 bits and ranks 2-9 at 3 bits (tier plan 4433333333, about 3.2 bits per expert weight). The 3 bits are the top three bits of the 4-bit store (MSB-3), so the SSD holds one copy. Ranks 0-1 add a 1-bit sidecar record (section 3.1).
- With every rank waited for, the tier plan passed all four teacher-forced fidelity targets on the real decoder (1.10x the 4-bit reference per corpus part): KLD code 0.00129, math 0.00298, IFEval 0.01763, multilingual 0.00989 [K, SB 'B12_NORM gate + dh_w4 on v3 tiers for the MAIN TRACK'; data BD:data/tfb12/v3ctl_c128_e8.npz].
- One-token decoding at 12 GB keeps the late rule: cold ranks 2-9 are added half a layer late (section 3.2). The verify loop waits for all ten ranks, so no rank is late in it [K, F12 'What the 12 GB exact loop (arm ex) is exact to'].
- **Memory.** The first 12 GB fit of this store had a lifetime peak footprint of 11.6 GB: one-token decoding, a code prompt, prefill included [K, SB 'TRUE peak footprint of the shipping configs', proc_pid_rusage lifetime maximum].
  - Earlier records quoted MLX's peak active memory, which reads 0.5-1.0 GB low, as that record notes itself.
  - We do not quote that run's speed. Its record gives decode speed only, from an older harness.

**The verified loop.**

- Today's loop drafts 15 tokens per check with a block drafter: our fine-tune of DFlash2 (Inco AI; Apache-2.0), at 4 bits ([DRAFTER]).
- It checks the drafts in calls of at most four rows. Each call reads and waits for every one of the top-10 ranks, and the check stops at the first miss [K, F12, same section].
- Section 3.3 describes the loop, section 2 says why it is verified and not exact, and table 6.1 gives its speeds.

**A wrong headline and a failed lossy mode (6-7 Nov).**

- Until 6 Nov, 40.8 tok/s was quoted as the 12 GB exact line.
  - It was the full-output speed of a two-pass arm (t80ck2), which commits rows certified by a cheap top-2 pass without the full check.
  - The decision map (section 4) had already scored that top-2 rule at theta 0.8 a code KILL (1.63 % flips) [K, F12 'CORRECTION for the shared plan'].
  - Section 8 keeps the withdrawn entry.
- Its successor had a top-1 first pass with a confidence gate (t80ck1u, theta 0.8).
  - It passed the map's code line by one flip, and was priced at 45.8 tok/s [P, SB 'Sat 04:2x: STEP 0 RESULT'].
  - It then failed HumanEval+ against its exact reference, and so did its theta 0.9 and 0.95 forms (sections 2 and 4.5).
- So the line's configuration today is the verified loop. Its lossy width rule passed only an IFBench test, on loose scoring (section 2).

**What we take from it** [G].

- Measure where the quality goes before choosing what to cut. Here it was the dropped experts, not the bits, and deferring them fixed most of it at no memory cost.
- A cheap screen orders candidates; it does not certify them. The map's code line passed a mode that the benchmark then failed (section 4.5). Since 8 Nov our quality gates use only the model's known benchmarks.
- A speed goes out with its label. The 40.8 was quoted as exact before anyone checked which arm had produced it.

### 3.1 Store and tiers (12 GB)

- Each token's ten experts are ranked by router weight. That gives tier plan 4433333333:
  - ranks 2-9 read only the 3-bit base record ("MSB-3", the top three bits of the 4-bit store);
  - ranks 0-1 also read a 1-bit sidecar record, which restores 4 bits.
  - The SSD holds one copy, and the pools hold base and sidecar slots separately.
- The embedding is trimmed to 4 bits, group 64.
- Its evidence is teacher-forced KLD on the real decoder; it predates the decision map (section 4).
- Printed settings [K, F12R:ex16 settings lines]:
  - tier 4433333333; C_m 47 base / C_l 32 sidecar slots per layer (9 % of the experts); pools 6.03 GB; MLX active 8.96 GB
  - drafter 0.928 GB of its own parameters
  - lifetime 10.583 GB after load

### 3.2 Pools and the loader

- A loader thread reads the missing experts of each layer with F_NOCACHE reads, which bypass the page cache, through a bounce buffer into pool slots. Reading straight into the slots was slower (section 8).
- The GPU posts each layer's routed ids to a shared buffer, and spin-waits on the loader's flags before the experts it needs.
- Python builds the next layer's graph while the GPU runs, with an async evaluation after each layer.
- **One-token decoding** defers cold ranks: ranks 2-9 that miss are added half a layer late, so their reads overlap the next half layer's compute. KTransformers (Chen et al., SOSP '25) published this as Expert Deferral (section 9).
- **The verify loop** waits for every rank ("ALLWAIT"). A verify call never defers, so its bits do not depend on the pools (section 2).

### 3.3 The 12 GB check loop

- **Drafts.** Our fine-tuned DFlash2 drafter, at 4 bits ([DRAFTER]), drafts 15 tokens per check.
- **Verify.** The 12 GB model verifies them in calls of up to 4 rows and stops at the first miss.
  - Code keeps 8.9-10.1 tokens per check; chat keeps 2.3-2.4 [K, F12R].
  - On chat, 40 % of checks keep a single token [K, F12C: chw13a].
- **PR1.** When a call keeps one row, the engine replays its commit at 2 rows instead of 1. This keeps the committed state off the width-1 path.
- **The tail rule** never leaves a 1-row last chunk inside a check: 4 + 1 becomes 3 + 2, and 2 + 1 becomes 3. The one remaining 1-row call, the cap's last token, waits for every rank (ALLWAIT).
- **P2'.** The first call reads 2 rows instead of 4 when the drafter's confidence in its first draft is below 0.9.
  - On chat it cuts SSD bytes per kept token 13 % and raises whole decode 9.6 % geo [K, F12 'T2 CHAT WIDTH stage 2'].
  - Code moves -1.0 %.

### 3.4 The 24 and 32 GB configurations

- **24 GB (w13b, as gated)** [K, BM:w24env; BM:w24d].
  - [MAIN]'s L + COH with 141 base and 26 sidecar expert slots per layer (14.87 GB of pools), no volatile expert tier, checks of at most 13 rows, the 4-bit DFlash2 drafter for every prompt type, and a trimmed prompt pass in 1,024-token chunks.
  - It replaced a 14-row variant under a rule set before the run: on the same 12 HumanEval prompts in one run, 130.3 against 125.9 tok/s pooled (+3.5 %), both under 24 GB there. One deviation

from that rule: the choice was confirmed at 01:33, after its 01:30 deadline; the comparison was re-derived from the raw rows [K, EV:RESULT_card_w24].
  - In a later same-run comparison on MBPP and chat, against an earlier variant with 130 base slots (w13t), it ran 91.2 vs 92.6 tok/s on MBPP, inside its own spread between two token-identical passes (2.9 %), and 30.6 vs 29.7 on chat [K, BM:w24f]. So the HumanEval gain that chose it does not carry over to MBPP.
- **24 GB (E24).** An older 24 GB stack [K, BM:WED 'E24']: stock kernels, the v3 tiers (4433333333), cache C 128, the MTP stream with FR-Spec + Scout drafting, and a wired-memory limit.
  - The main 45 GB engine cannot go below cache C ~160 and stay exact, because its 16-row checks need up to 160 experts resident per layer call. That is why the exact 24 GB configuration uses this stack; w13b caps its checks at 13 rows and is lossy.
- **32 GB (C32 cell).** Cache C 160, 12 IFEval, one pass per arm, quiet [K, BM:c32].
  - e32: the exact stack.
  - zs32: M1-B + COH + SB1, lossy.
- Against cache C 256 on the same prompts: E drops 7 % (36.4 to 33.8) and zs 1 % (51.9 to 51.3). zs's cold-rank skips absorb most of the smaller cache: misses per check are 42.7 vs 24.6, while E's are 168.4 vs 89.8 [K, BM:WED 'C32']. [RELEASE] section 2 gives both configurations.

# 4. The decision map

The decision map screens a lossy rule in about four minutes of GPU time, before any benchmark run. This section says what it measures, how it was validated, what it screened, and how its verdicts compared with the benchmarks.

## 4.1 What it measures

- **Flips.** The map is a teacher-forced harness on the 125B.
  - It replays the model's own greedy answers with one part of the computation changed. It counts how often the top-1 token changes (a "flip") against the unchanged model.
  - Flips are counted per domain, and per confidence class of the unchanged model: sure (p1 >= 0.99), tie (p1 - p2 <= 0.1) and mid (the rest).
  - Non-tie flips (sure + mid rows) are given in brackets [DM:dm_common.py, header].
- **Windows.** 12 code (HumanEval), 12 IFEval and 12 multilingual prompts, each with the model's greedy answer. That is 10,316 answer rows: 3,366 / 3,397 / 3,553 per domain [K, BD:data/papers/dm_out/gate.txt].
- **The unchanged model.**
  - It reads every routed expert at 4 bits, from the original 4-bit store, with 51 expert slots per layer and an exact prefill. The pool size changes only the map's own speed.
  - This is not the 3.2-bit store our engines run. Map flips are therefore measured against the 4-bit model (4.6).
  - Per answer row, the map keeps the top-8 ids and log-probs [DM:dm_common.py, configuration note 'A1' and `measure`].
- **Rules.** A rule acts on the router's rank-sorted choices.
  - A keep mask per (row, layer, rank) zeroes a dropped expert's weight, without renormalizing.
  - A 3-bit form reads the top three bits of the same records.
- **Masks** come from CPU replays of the engine's per-layer expert pool [NI:scripts/dm_oea_masks.py, header; unchanged copy BD:scripts/m1/dm_oea_masks.py]:
  - LRU per layer, carried across a domain's windows and warmed by the prompt's rows.
  - Each verify check is cut where the drafter's running confidence (the product of its drafts' confidences) falls below 0.3, as the 45 GB engine does, and loads its union of experts.
- **Plain LRU is slightly pessimistic.** The engine adds a popularity prior. The replay's base-pool misses run 1.05-1.25x the engine's pool model, so the masks drop slightly more than the engine would [NIQ 'Tue 17:01: PRE-REGISTRATION, Q1-ON-MISS ON LIVE-CUT CHECKS'].

## 4.2 Validation

- **Wiring** [K]: the unchanged run reproduces the top-1 of an independent extraction on 10,316 of 10,316 rows (100.000 %) [BD:data/papers/dm_out/gate.txt].
- **Reproducibility** [K]: on 8 Nov the map was re-run through a new code path, with store and architecture

switches added for the next model. The results were identical to the stored ones [BD:data/q4/q4_repro_check.out]:
  - the run output was byte-equal to the 2 Nov baseline;
  - the routing ids were equal on 36 of 36 windows;
  - the captured expert inputs and weights were equal on 3 of 3 windows.
- **Cost** [K]:
  - one 36-window arm takes 229-239 s of GPU time [BD:data/m1/chat_gates_24_12_arms.log];
  - the masks for three arms take 45 s on one CPU core [BD:data/m1/masks/build_c51.log].

### 4.3 Pass lines

- **Lines per domain**, all-row flips [non-tie flips]: code 1.34 [0.87] %, IFEval 5.42 [2.22] %, multilingual 3.69 [1.63] %.
- **Where they come from.** They are the flip rates of an earlier lossy rule: the top-2 experts on rows where the drafter is at least 0.99 sure.
  - Re-derived from that arm's stored output: code 45 / 3,366, IFEval 184 / 3,397 and multilingual 131 / 3,553 all-row flips [K, BD:data/m1/lines_check_gate_top2.out].
  - The all-row lines were pre-registered with PASS at or under 1x, BETWEEN up to 1.5x and KILL above [NIQ 'Tue 10:16: MAP RESULT', 3 Nov].
  - A PASS now also needs the non-tie flips at or under the bracketed line (BD:scripts/m1/m1_quality.py, `LINES`).
- **So a PASS means "no more flips than that rule"**, not a benchmark tolerance.
  - We found no benchmark verdict for that rule itself.
  - The benchmark run planned for it replaced it with a wider form before running [K, EV:PREREG_relax, revision 3].
- Two arms scored on the same rows are compared by a paired McNemar test (flips only in A against flips only in B).

### 4.4 What it screened

**QOM** ("Q1 on miss") is the family most of the byte cuts below use.

- On rows where the drafter is at least cf sure, QOM keeps ranks 0-1, the check's union members, and every rank 2-9 expert already resident after the check's loads. It drops only the rank 2-9 experts that would miss [NIQ 'Tue 17:01'].
- At cf 0, every drafted row qualifies. Below, a row is "cut" when a rule drops experts from it.
- The last row of a check verifies the next draft, or no draft at all (the bonus row). It stays exact unless the arm says otherwise. Two kinds of last-row cut were screened:
  - QOM's own form ("last row cut"): the row keeps its resident rank 2-9 experts too [NI:scripts/dm_oea_masks.py, `_bq` / `_ball`];
  - the harder bonus-row form ("bonus row cut"): the row keeps only ranks 0-1 and the check's union members. It reads the same bytes as QOM's form, because resident experts cost no reads, but it drops more, so a PASS for it is also a PASS for QOM's form [BD:scripts/m1/bytes10_bonus_masks_bt.py, header; a copy with only its folders changed].

**At the 45 GB pool.** These are the engine's per-layer pool sizes, mean 295 base slots of 512. Cells are all-row % [non-tie %] [K, BD:data/m1/m1_quality.out; the first two rows BD:data/m1/lines_check_gate_top2.out].

| rule | map arm | code | IFEval | multilingual | map verdict |
|---|---|---|---|---|---|
| top-2 on rows >= 0.99 sure (sets the lines) | gate_top2_cf099 | 1.34 [0.87] | 5.42 [2.22] | 3.69 [1.63] | the line |
| top-2 on rows >= 0.9 sure | gate_top2_cf09 | 1.78 [1.21] | 8.86 [5.15] | 7.60 [4.93] | FAIL |
| QOM cf 0.9 | qom_0.9_live | 0.48 [0.03] | 2.24 [0.19] | 1.44 [0.21] | PASS |
| QOM cf 0.5 | qom_0.5_live | 0.36 [0.06] | 2.68 [0.48] | 1.63 [0.24] | PASS |
| QOM cf 0 | qom_0_live | 0.42 [0.06] | 2.53 [0.45] | 1.77 [0.33] | PASS |
| QOM cf 0.5, bonus row cut when the next draft is >= 0.5 sure | qom_0.5_live_bonus5g | 0.48 [0.12] | 2.88 [0.55] | 1.91 [0.39] | PASS |
| QOM cf 0, every bonus row cut | qom_0_live_bonus | 0.65 [0.24] | 6.12 [2.77] | 6.50 [3.93] | FAIL (chat) |
| QOM cf 0, a last row cut when it verifies a draft | qom_0_live_bq | 0.39 [0.15] | 4.09 [1.19] | 3.26 [0.91] | PASS |
| same, every last row cut | qom_0_live_ball | 0.51 [0.21] | 4.03 [1.00] | 3.26 [0.91] | PASS |
| QOM cf 0, sure rows' ranks 0-1 at 3 bits | qom_0_live_t3 | 0.71 [0.21] | 4.45 [1.71] | 4.14 [1.69] | FAIL (multilingual) |

| rule | map arm | code | IFEval | multilingual | map verdict |
|---|---|---|---|---|---|
| top-8 experts on every row | top_k8 | 0.65 [0.21] | 5.24 [1.90] | 4.31 [1.54] | FAIL (multilingual) |
| top-2 + the check's union on rows >= 0.9 sure | q1u_0.9_live | 1.25 [0.69] | 6.45 [2.90] | 3.77 [1.51] | FAIL (chat) |
| every row's top-2, at 3 bits | t3_top2_all | 4.49 [3.68] | 27.35 [23.56] | 31.75 [28.21] | FAIL |

**At the 24 and 12 GB pools** (8 Nov, the last two rows 10 Nov; information, see 4.6) [K, BD:data/m1/chat_gates_24_12.out; the last two rows BD:data/m1/SCREEN_read_qom_0_live_c47.txt and BD:data/m1/SCREEN_read_q1u_0_live.txt]:

| rule | map arm | expert slots per layer | code | IFEval | multilingual | map verdict |
|---|---|---|---|---|---|---|
| QOM cf 0.5, bonus row cut when the next draft is >= 0.5 sure (the 24 GB chat route) | qom_0.5_live_c147_bonus5g | 147 | 0.71 [0.33] | 4.30 [0.93] | 2.73 [0.85] | PASS |
| QOM cf 0 | qom_0_live_c147 | 147 | 0.74 [0.24] | 4.50 [1.26] | 3.26 [0.76] | PASS |
| QOM cf 0, every last row cut | qom_0_live_ball_c147 | 147 | 0.92 [0.54] | 6.36 [2.96] | 4.11 [1.85] | FAIL (chat) |
| QOM cf 0 | qom_0_live_c51 | 51 | 1.16 [0.54] | 8.80 [4.96] | 5.26 [2.66] | FAIL (chat) |
| QOM cf 0, every last row cut | qom_0_live_ball_c51 | 51 | 1.40 [0.93] | 10.57 [6.53] | 7.99 [5.02] | FAIL |
| QOM cf 0 | qom_0_live_c47 | 47 (the 12 GB engine's own pool) | 1.37 [0.69] | 9.24 [5.54] | 5.71 [2.94] | FAIL (code by one flip, and chat) |
| top-2 + the check's union on every drafted row, no resident ranks kept | q1u_0_live | any (keeps no resident ranks) | 1.96 [1.36] | 15.72 [12.13] | 11.15 [8.02] | FAIL |

**Reading** [G from the [K] rows]:

- At the 45 GB pool, the QOM family stays well under every line. Cutting the last row of each check adds about 1.5 points of IFEval flips and still passes.
- 3-bit sure rows and top-8 fail on multilingual.
- The same rules flip more as the pool shrinks, because more experts miss and are dropped:
  - at 147 slots (24 GB), the cf 0.5 route and plain cf 0 still pass, but cutting every last row fails on chat;
  - at 51 slots (12 GB), even plain cf 0 fails on chat; at 47, the 12 GB engine's own pool, it also fails code, by one flip (46 of 3,366 rows against 45 allowed).
- Keeping only the top-2 and the check's union on every drafted row, without the resident ranks, fails every line.

## 4.5 The map against the known benchmarks

Two rules have both a map score and a benchmark verdict.

**Code: the 12 GB two-pass FAST family.**

- On the map, all three forms were under the code line [K, SB 'Sat 04:2x: STEP 0 RESULT']:

| form | all-row flips | non-tie flips |
|---|---|---|
| theta 0.8 | 1.31 % | 0.78 % |
| theta 0.9 | 1.16 % | 0.66 % |
| theta 0.95 | 1.10 % | 0.60 % |

- Theta 0.8 passed by one flip. Its record said that a second sample or the benchmark could tip it.
- On HumanEval+ against the 12 GB exact reference (REF2-T), 60 prompts, both prompt orders, all three FAILED [K, EV:RESULT_t80ck1u]:

| form | pass rate vs REF2-T's 95.0 % | delta | p |
|---|---|---|---|
| theta 0.8 | 83.3 % | -11.67 pt | 0.0156 |

| form | pass rate vs REF2-T's 95.0 % | delta | p |
|---|---|---|---|
| theta 0.9 | 86.7 % | -8.33 pt | 0.0625 |
| theta 0.95 | 90.0 % | -5.00 pt (at the bar) | 0.25 |

- Every discordant task went against the arm.
- Against the 45 GB exact reference, theta 0.95 passes (-1.67 pt, p 1.0).
- **So a map PASS on code did not predict the benchmark** for this first-pass form.

**Chat: the 45 GB rule M1-B.**

- M1-B applies QOM at cf 0 to the drafted rows of the chat sets, and gates the bonus row of each check as well.
  - Its 3-bit part skips only the gated rows' cold 1-bit sidecars. A pair whose sidecar is resident still runs at 4 bits.
  - It sits on top of L: the certified 45 GB configuration plus the exact-safe bundle of six switches, which is bitwise in the exact mode ([MAIN] 6.1).
  - Sources: SB 'Sun 04:2x: M1-B [K] AGAINST MY [P], AND WHAT THE ENGINE'S M1-B IS'; EV:RESULT_chat_m1b, 'The config'.
- **On IFEval** against the 45 GB exact reference (NAX6-N), it gave the same scores [K, EV:RESULT_chat_m1b]:
  - forward order: 86.0 vs 87.0 % (-1.00 pt, p 1.0);
  - reverse order: 83.0 vs 87.0 % (-4.00, p 0.388).
- **The map never scored M1-B itself.** It brackets the QOM part [K, BD:data/m1/m1_quality.out]:
  - every last row cut, no 3 bits (qom_0_live_ball): IFEval 4.03 [1.00], multilingual 3.26 [0.91], PASS;
  - last row cut when it verifies a draft, and every sure row's ranks 0-1 at 3 bits (qom_0_live_bq_t3): IFEval 6.09 [2.48], multilingual 5.29 [2.36], FAIL.
  - The engine's cold-sidecar rule should sit near the first [G]. The IFEval result agrees.
- **Multilingual is information; the model card's benchmarks decide.**
  - Against NAX6-N, M1-B's own multilingual output deviates on 15.29 % of 7,033 tokens (334 blunders per 10K) [K, EV:RESULT_chat_m1b].
  - The map cannot explain that number. Its arms change only the QOM part, on the 4-bit base, while M1-B also carries L's other levers.
  - The deviation figure is information under the benchmark-only rule. The MGSM line registered in its place is on hold: the engine types MGSM's prompts as math, so they would not exercise M1-B's chat lever [EV:PREREG_chat_gate, amendment 3; [EVAL]].
  - M1-B + COH's gate is now the model card's benchmarks: IFBench, LiveCodeBench v6 and GPQA Diamond, both orders, against NAX6-N [EV:PREREG_card_gates, arm m1b]. They do exercise the chat lever: the engine types 141 of the 150 IFBench gate prompts and 67 of the 100 GPQA Diamond ones as chat. It passed on 9 November: M1-B + COH is "lossy, same scores on IFEval, IFBench, LiveCodeBench v6 and GPQA Diamond (both orders)" [K, EV:RESULT_card_m1b].

### 4.6 Limits

- **Teacher-forced.** A changed token never feeds the next one, so the map cannot see errors that compound in free decoding.
- **Small.** 36 windows, about 3,400 rows per domain. Flips are top-1 changes, not task scores.
- **Another base.** The map's unchanged model reads every expert at 4 bits. Our engines run the 3.2-bit store, and their exact references (NAX6-N, REF2-T) run it too. A map arm and its engine form therefore apply the same rule to different base bits. The map has no arm for the tier plan itself.
- **Relative lines.** The lines are relative to an earlier lossy rule. Since 8 Nov our quality gates are scored only on the model's known benchmarks: lossy against its exact reference, on the same harness [EV:RESULT_chat_m1b, header]. The map orders candidates and rejects bad ones in minutes; its verdicts are information.
- **Pool model.** The masks model the engine's pool with plain LRU (4.1).

## 5. BYTES10: SSD bytes per kept token

### 5.1 The metric and the target

- **The metric:** SSD bytes read per kept (accepted) token.

  - Bytes are priced on the union of each verify check: an expert read once serves every row of the check.
  - A miss is a routed expert that is not resident when its check needs it.
- **The target.** BYTES10 is the target of at most about 10 MB per kept token, first at 24 GB, then at 12 GB [FG:c5].
  - At the measured ~12 GB/s (section 1.1), 150 tok/s of chat needs at most about 80 MB per kept token [P].
  - 10 MB per kept token would leave the SSD more than 1,000 tok/s of headroom [P].

## 5.2 The replay

- **What it replays.** A CPU replay of the map's 24 chat windows (12 IFEval, 12 multilingual), with each check cut at the drafter's running confidence 0.3 (4.1). It keeps per-layer pools of base and sidecar slots at each memory line:
  - eviction is LRU, or the offline optimum (OPT, Belady), which no engine can reach;
  - 3-bit base records and 1-bit sidecars are priced from their record sizes: 2,162,688 B per base record and 770,048 B per sidecar.
- **Pools**, base + sidecar slots per layer:
  - 45 GB: the engine's per-layer sizes (means 295 + 89);
  - 24 GB: 147 + 46;
  - 12 GB: 51 + 32.
- **Code:** BD:scripts/m1/bytes10_core.py (an unchanged copy of the published replay; constants `MR`, `LR`, `POOLS`) and BD:scripts/m1/m1_replay.py. Output headers restate the pools and record sizes [BD:data/m1/m1_replay*.out].

## 5.3 Levers, priced

Cells are MB per kept token, IFEval / multilingual, LRU unless named [P, BD:data/m1/m1_replay.out (45 GB), m1_replay_24.out, m1_replay_12.out]. Map verdicts are information (4.6) [K, BD:data/m1/m1_quality.out (45 GB), chat_gates_24_12.out (147 / 51 slots)].

| rule | 45 GB | 24 GB | 12 GB | map verdict (arm) |
|---|---|---|---|---|
| exact (every routed rank read) | 38.1 / 33.2 | 210.2 / 182.6 | 621.6 / 486.6 | reference |
| same, offline optimum (OPT) | 19.6 / 19.7 | 93.9 / 83.5 | 386.8 / 304.4 | (a bound) |
| top-8 on every row | 30.2 / 25.9 | 153.1 / 127.7 | 466.2 / 356.5 | 45 GB: FAIL multilingual (top_k8) |
| no sidecar reads (ranks 0-1 at 3 bits when cold, U3) | 26.6 / 24.7 | 181.6 / 161.7 | 584.0 / 459.4 | worst case on the map: FAIL multilingual (qom_0_live_t3) |
| QOM cf 0, bonus row exact | 25.7 / 24.3 | 105.1 / 95.5 | 275.7 / 242.0 | PASS at 45 GB and at 147 slots; FAIL at 51 |
| QOM cf 0.5, bonus row cut when the next draft is >= 0.5 sure | 22.8 / 22.0 | 81.7 / 77.0 | 216.2 / 199.4 | PASS at 45 GB and at 147 slots (the 24 GB chat route) |
| QOM cf 0, every last row cut | 16.0 / 13.0 | 48.6 / 35.2 | 113.3 / 82.7 | PASS at 45 GB; FAIL at 147 and at 51 |
| same, OPT | 9.0 / 8.6 | 21.9 / 18.4 | 55.0 / 40.8 | (a bound) |
| same, and no sidecar reads at all | 4.5 / 4.4 | 20.0 / 14.2 | 75.7 / 55.5 | FAIL (45 GB qom_0_live_bq_t3; 147 slots qom_0_live_ball_c147_t3) |

- **The exact rule at 24 and 12 GB** reads 5.5x and 16x the 45 GB bytes on IFEval [P].
- **The 24 GB chat route** cuts this by 61 % (81.7 vs 210.2) [P], and passes the map at 147 slots. No engine has run it at 24 GB.
- **Moving base slots to sidecar slots at equal bytes** (S base slots become 2.81 S sidecar slots per layer) [P]:
  - the exact rule: about flat at 45 GB (16 slots: -1 % / +1 %), and worse beyond;
  - QOM cf 0 with every last row cut: -41 % / -35 % at 45 GB (32 slots: 9.4 / 8.4), and -30 % / -29 % at 24 GB (32 slots: 34.1 / 25.1). None at 12 GB (best -2 % / -1 %).
  - The engine did not confirm the 45 GB split (5.5).

## 5.4 The engine at 45 GB

The byte cuts were first measured on the 45 GB engine ([MAIN] covers that line).

- Source: BM:chs2. 12 IFEval prompts, 256 base slots per layer, one pass per arm, quiet machine.

- The report prints whole decode. Whole request and first token are re-derived from the same run records (first token = the record's pfs, as in [MAIN]'s harness) [K, BD:data/papers/chat45_rows_check.txt]. The table compares bytes and is not a speed headline.

| arm | label | whole decode tok/s pooled | whole request tok/s pooled | first token s, median (min-max) | misses per check | MB per kept token | lifetime peak GB |
|---|---|---|---|---|---|---|---|
| L = the certified configuration + the exact-safe bundle (first run) | lossy, not gated by itself (below) | 44.8 | 20.4 | 3.05 (2.60-3.85) | 97.8 | 41.8 | 42.37 |
| L (repeat, last run) | same | 43.7 | 20.8 | 3.02 (2.57-3.80) | 91.1 | 40.9 | 42.41 |
| L + QOM on drafted rows, bonus row exact | lossy, ungated | 45.3 | 21.3 | 2.94 (2.50-3.72) | 42.8 | 22.8 | 42.38 |
| M1-B (+ the gated last row) | lossy, not gated by itself; with COH it is the chat configuration below | 49.0 | 23.0 | 2.96 (2.40-3.74) | 25.8 | 15.5 | 42.38 |
| M1-B + pool split | lossy, ungated | 52.0 | 22.1 | 3.10 (2.56-3.84) | 28.2 | 15.1 | 42.33 |
| E, the exact stack, before coherent reads | not exact on chat without them: tokens equal one-token decoding on 12 / 12 prompts here, 3 / 12 in a later run ([MAIN] 6.1a) | 36.6 | 20.7 | 2.97 (2.25-3.70) | 89.8 | 40.0 | 41.70 |

- **Labels at 45 GB.** L is the certified configuration ("v1' + wired") plus the exact-safe bundle. The bundle is bitwise only in the exact mode, so the certificate does not carry to L ([MAIN] 6.1c). These runs also predate coherent reads (COH), which change chat tokens on the one-call path ([MAIN] 6.1). The gates that cover chat ran with COH: M1-B + COH below, and L + COH, the published 45 GB configuration, which passed the model card's IFBench, GPQA Diamond and LiveCodeBench v6 in both orders against NAX6-N on 9 November [K, EV:RESULT_card_lcoh].
- **M1-B against L** [P from the [K] rows]: +10.7 % whole decode against the mean of the two L runs (49.0 vs 44.25 tok/s). It reads 63 % fewer MB per kept token (15.5 vs 41.35), and has 73 % fewer misses per check (25.8 vs 94.45).
- **M1-B + split.** Its further +6 % comes with more kept tokens per check (2.72 vs 2.41; the lossy outputs differ). It does not come from fewer bytes (15.1 vs 15.5) [K, same source]. On whole request it trails M1-B (22.1 vs 23.0 tok/s): its outputs are shorter (1,480 vs 1,580 tokens) and its first tokens slower (38.5 vs 36.5 s in total) [K, BD:data/papers/chat45_rows_check.txt].
- **The chat configuration: M1-B + COH (coherent pool reads)** [K, BM:chs5 arms z0 and z0b, BM:ttft]. 12 IFEval prompts, cap 256, 256 base slots, quiet machine, two passes pooled:
  - whole decode 49.8 tok/s (passes 49.7 and 49.9);
  - whole request 24.4 tok/s;
  - first token median 2.92 s (2.34-3.81, 24 runs);
  - lifetime peak 42.38 GB (passes 42.36 and 42.38);
  - 24.6 misses per check; 14.3 and 13.8 MB per kept token.
- **Label:** lossy, same scores on IFEval, IFBench, LiveCodeBench v6 and GPQA Diamond (both prompt orders, against NAX6-N) (4.5).
- **The speed run and the gate ran the same configuration** [K]:
  - the same keys (BM:git 275c592, arm z0, against EV:RESULT_chat_m1b's config);
  - frozen code trees (snapshots 65cd673 and 342f914) identical file for file, apart from each snapshot's own folder name [BD:data/papers/snapshot_chs4_vs_chs5.txt];
  - the compiled extension was not compared.
- **First token is about 2.9 s at 45 GB without PF-D**, because the prefill reads the prompt's experts from the SSD. On these prompts it is half of the whole request: first tokens total 36.4 s of 71.4 s on the first pass and 36.6 s of 71.5 s on the second [K, BM:ttft].
- **Chat as shipped** adds SB1 (the 1-row check) and PF-D v4. PF-D builds a short prompt's routed experts from the copies already in memory and reads only the missing parts, so prefill reads 9.6 GB from the SSD per chat prompt instead of 20.4 GB [K, BM:pfd]. It follows Strata's prompt-path design; no Strata code is included ([MAIN] section 4.5). Both additions keep this configuration's tokens on 12 / 12 prompts ([EVAL] Table 1b). One pass, same label: first token 2.26 s (1.55-2.59), whole decode 50.7 tok/s, whole request 28.8 tok/s, lifetime peak 43.69 GB [K, BM:pfd, arm pdz4a].

## 5.5 Replay against the engine

- **Exact bytes match.**

  - At 45 GB, the replay's exact rule reads 38.1 MB per kept token on IFEval against the engine's exact arm at 40.0 [K, BM:chs2]: within 5 %.
  - At 12 GB, it reads 621.6 [P] against the verified loop's measured 556 MB per kept token on chat [K, F12R:chat12]: +12 %.
  - The prompts are the same IFEval set, but the checks differ. The replay uses the map's windows and the 45 GB drafter's cut. The 12 GB loop drafts 15 tokens and checks them in 4-row calls, with 47 base slots per layer, not 51.
- **QOM's cut: the replay brackets the engine.**
  - Misses per check [P, BD:data/m1/m1_replay.out]: the replay predicts -38 % for QOM cf 0 with the last row cut, with no 3 bits (63.3 to 39.5; the sidecar misses unchanged). With no sidecar reads at all, it predicts -92 % (4.8).
  - The engine's M1-B, whose 3-bit part skips only cold sidecars, measured -74 % [K] (97.8 to 25.8).
  - Bytes per kept token: the replay's -58 % (16.0 vs 38.1) [P] against the engine's -63 % (15.5 vs 41.8) [K]. Source: SB 'Sun 04:2x: M1-B [K] AGAINST MY [P]'.
- **The pool split did not carry over.**
  - The replay priced 41 % fewer bytes with 32 base slots per layer moved to sidecars [P]. The engine measured -3 % (15.5 to 15.1) [K].
  - The engine already holds more sidecar slots (112 + 32 per layer) than the replay's 45 GB line (mean 89), so most of that lever was already in place [G from the configs; same SB section].
- **Rule kept:** a priced lever is quoted as a result only after an engine A/B.

### 5.6 After the bytes, the GPU check

- **At 45 GB the check now costs more than the reads.** At M1-B's ~25 misses per check, the chat check still takes 40.0-40.5 ms of a 48.5-48.8 ms cycle. Only 5.7-6.3 ms of it is SSD service time [K]. With 2.4 kept tokens per check, that is the measured ~50 tok/s.
  - Sources: BM:chs2 arm x2qm2 (check 40.5 ms, cycle 48.8, service 6.3, kept 2.41) and BM:chs5 arm z0 (check 40.0, cycle 48.5, service 5.7, kept 2.43).
  - An earlier per-check analysis of the 45 GB chat loop found the same: the check costs ~40 ms in the lossy configuration even at 1-2 rows, and SSD misses add ~4 ms at 1-2 rows [K, FG:chatbound].
- **A ceiling** [P]:
  - With no SSD wait at all, a check would still take at least ~34 ms (40.0 - 5.7, if all of the service time is on the critical path).
  - Chat at 45 GB then tops out near 2.43 / (34.3 + 8.5 ms of drafting) ≈ 57 tok/s, unless the check gets faster or more tokens are kept per check.
- **At 24 and 12 GB, the SSD alone caps the speed** [P: ~12 GB/s ÷ the MB per kept token of 5.3]:

| line | exact rule | the chat route (QOM cf 0.5, gated bonus row) |
|---|---|---|
| 24 GB | ~57 tok/s | ~147 tok/s |
| 12 GB | ~19 tok/s | ~55 tok/s |

- These are bounds, not predictions. No engine has run the route at 24 or 12 GB, and the check's own cost comes on top.
- At 12 GB the check's fixed cost is 42-59 ms per call (section 7).

## 6. Results

### 6.1 The table

- **Definitions** ([MAIN] section 2, with one note on our harness).
  - Whole decode = generated tokens / the decode window. The window starts after prefill and holds the drafter's setup, check 0 and all host work.
  - Whole request = generated tokens / (prefill + decode window), the request's wall time.
  - First token = prefill + the decode window up to the end of check 0, which commits the first output tokens. Our prefill timer covers the prompt minus its last token; the last token goes through check 0.
- **Derivation.** All 12 GB rows are derived in one way from the raw logs (F12R), and each agrees with its run's own report.
- **Workloads.** Greedy decoding with an EOS stop; cap 256 unless marked.

| line | config | label | prompt set, cap | whole decode tok/s pooled (geo) | whole request tok/s pooled (geo) | first token s median (min-max) | lifetime peak GB | mark | source |
|---|---|---|---|---|---|---|---|---|---|
| 12 GB | verified loop | verified | 12 HumanEval + 12 MBPP | 18.34 (18.38) | 13.12 (12.44) | 4.29 (3.47-5.85) | 11.140 | K, quiet | F12R:tab12e |
| 12 GB | verified loop | verified | 12 IFEval | 12.76 (12.88) | 10.09 (9.02) | 2.92 (2.33-3.71) | 10.962 | K, quiet | F12R:chat12 |
| 12 GB | two-pass FAST, theta 0.95 | lossy, failed its gate | 12 HumanEval + 12 MBPP | 27.90 (27.58) | 17.60 (16.21) | 4.15 (3.29-5.71) | 11.280 | K, quiet | F12R:tab12f |
| 12 GB | two-pass FAST, theta 0.8 | lossy, failed its gate | 16 HumanEval only | 44.34 (46.33) | 24.92 (24.49) | 4.30 (3.61-5.18) | 11.245 | K, quiet; HumanEval only | F12R:c1u16 |
| 12 GB | verified loop | verified | 16 HumanEval only | 18.47 (18.42) | 13.90 (13.66) | 4.51 (4.21-5.93) | 11.144 | K, quiet; HumanEval only | F12R:ex16 |
| 12 GB | verified + PR1 + tail (A') | verified | 12 IFEval, cap 128 | 12.56 (12.68) | 9.32 (8.57) | 2.90 (2.37-3.65) | 10.962 | K, share lock (rough) | F12R:chw13a |
| 12 GB | A' + width rule (P2') | lossy (rounding), same scores on IFBench loose (one order); strict lower | 12 IFEval, cap 128 | 13.97 (13.89) | 10.07 (9.12) | 3.04 (2.32-3.70) | 11.012 | K, share lock (rough) | F12R:chw14c2 |
| 12 GB | A' | verified | 16 HumanEval only | 18.69 (18.68) | 14.11 (13.87) | 4.48 (4.22-5.64) | 11.145 | K, share lock (rough) | F12R:chw14ha |
| 12 GB | P2' | lossy (rounding), same scores on IFBench loose (one order); strict lower | 16 HumanEval only | 18.47 (18.50) | 13.98 (13.76) | 4.48 (4.20-5.66) | 11.159 | K, share lock (rough) | F12R:chw14h2 |
| 24 GB | w13b: L + COH, 141 + 26 slots, rows <= 13 | lossy, same scores on IFBench, GPQA Diamond and LiveCodeBench v6 (both orders) | 12 HumanEval, 256 | 130.33 (geo 130.3; passes 130.2, 130.5) | 40.8 | 3.54 (3.20-4.78) | 23.82 | K, quiet | BM:w24d (arms w13e, w13f) |
| 24 GB | w13b | same | 12 MBPP, 256 | 91.25 (geo 93.7; passes 90.0, 92.6) | 26.1 | 3.34 (3.09-4.62) | 23.74 | K, quiet | BM:w24f (arms bf1, bf2) |
| 24 GB | w13b | same | 12 IFEval, 256 | 30.58 (geo 28.7; passes 30.1, 31.1) | 18.1 | 3.09 (2.45-4.16) | 23.66 | K, quiet | BM:w24f (arms bf1, bf2) |
| 24 GB | w13b | same | 12 math (MATH-500), 256 | 40.76 (geo 40.9; passes 40.7, 40.9) | 27.4 | 2.86 (2.55-4.92) | 24.01 (over 24.0 on a 520-token prompt) | K, quiet | BM:w24g (arms bg1, bg2) |
| 24 GB | E24 | exact to its own stock-kernel reference | 12 HumanEval, no EOS stop | 41.07 (geo 41.28; passes 41.65, 40.50) | not recorded (an older end-to-end definition prints 26.5) | not recorded | 23.86 | K, quiet | BM:e24 per-prompt lines; NI:NUMBER_CHECK 4 |
| 24 GB | E24m | same | 12 MBPP, EOS stop | 37.43 (geo 38.36; passes 37.37, 37.49) | not recorded (older definition: 18.6) | not recorded | 23.52 | K, quiet | BM:e24m per-prompt lines; same |
| 32 GB | e32: config E at cache C 160 | exact to NAX6-N (config E, [MAIN]); this run: tokens 12 / 12, logits not hashed | 12 IFEval, 256 | 33.81 (34.5) | 19.82 | 2.83 (2.24-3.62) | 30.62 | K, quiet | BM:r57 e32_160_runs.json; NI:NUMBER_CHECK 0a-2 |
| 32 GB | zs32: M1-B + COH + SB1 at C 160 | lossy, not gated at 32 GB (its IFEval gate ran at C 256) | 12 IFEval, 256 | 51.33 (54.0) | 22.52 | 2.99 (2.47-3.84) | 31.22 | K, quiet | BM:r57 zs32_160_runs.json; same |

- **Notes on the rows.**
  - HumanEval flatters, because its answers copy the prompt's docstring, so the 16-HumanEval rows

are secondary.
  - E24's HumanEval run did not stop at EOS: it decoded post-answer text to 256 tokens. Its MBPP row is the EOS-stop re-run, E24m.
  - The 24 GB report predates our standard metrics ([MAIN] section 2). Its "end to end" column is an older, third definition (below both the pooled whole request and the mean of per-prompt rates), so whole request and first token are "not recorded" there.
  - w13b's HumanEval row pools W24d's two passes (w13e, w13f); its MBPP and IFEval rows pool W24f's two passes (bf1, bf2), a later run, and its math row W24g's (bg1, bg2), in which every answer ran to the 256-token cap.
  - Missing at 12 GB: math (MATH-500), and MBPP for the width rule. No new runs were made for the paper (section 10).

## 6.2 Code against chat (12 GB)

- **Code keeps 8.9-10.1 tokens per check; chat keeps 2.3-2.4** [K, F12R]. The block drafter guesses code well, because answers echo the prompt, and chat badly.
- **Each check still reads its first call's rows.** At about 0.3 GB of SSD per row [K, F12P: 0.61 GB per 2-row first call], chat reads 491-564 MB per kept token against code's 436-441 [K, F12R]. The chat whole decode is two-thirds of code's.
- **The two-pass FAST mode** makes code faster (x1.52 on 12 HumanEval + 12 MBPP) but failed its gate (section 2).

## 6.3 Memory

- **Every 12 GB row peaks at 10.96-11.28 GB** [K, F12R]. The pools are 6.03 GB, and after load the process is at 10.58 GB.
- **Decode adds 0.4-0.7 GB** above the after-load 10.583 GB [K, F12R]. We read it as MLX's working buffers and the drafter's state [G].
- **Strata's smallest setting** puts 10.8 GB in everything but its expert arena, so it peaks at 14.19 GB with a 3.37 GB arena [K, FG:low].
- **24 GB (w13b): under 24 GB in our runs on prompts of up to about 500 tokens, with little room; longer prompts use more, 28.2 GB at 30K tokens** [K].
  - On the 12 HumanEval prompts (103-367 tokens) it peaks at 23.82 and 23.78 GB in its two passes [K, BM:w24d], and on 12 MBPP and 12 IFEval prompts (37-237 tokens) at 23.66-23.74 GB [K, BM:w24f]. On 12 math prompts (50-520 tokens) it reached 24.01 GB, 14 MB over the line, in the prompt pass of the 520-token prompt, the ninth of the 12 measured prompts (two warm-ups ran first), in both passes [K, BM:w24g].
  - Its quality gate made 764 prompt runs of IFBench, GPQA Diamond and LiveCodeBench v6 in 40 processes. 22 processes went over 24.0 GB, at most to 25.112 GB on the gate's longest prompt (2,801 tokens). All 14 IFBench-only processes stayed under; their prompts have at most 206 tokens [K, EV:RESULT_card_w24].
  - Longer prompts use more: 25.03 GB at 3,003 tokens, 25.67 GB at 6,980, 25.82 GB at 9,988 and 28.17 GB at 29,988 (default prefill; 256-token answers at 3,003 and 6,980 tokens, 64-token ones at 9,988 and 29,988). The 3,003-token run had a 36 GB Mac simulated on this one: memory locked away left the engine about 25 GB, and its GPU working set was capped at 27 GB [K, LP:data/tue_afternoon_1110.md sections 1, 2 and 4].
  - Per prompt, in the gate no process crossed 24.0 GB on a prompt of 560 tokens or fewer (the shortest at a crossing had 561 tokens), but the 520-token math prompt above crossed by 14 MB, and a 448-token GPQA prompt raised its process's peak to 23.954 GB, 46 MB under the line. Hence "up to about 500 tokens", as measured, not as a bound: the same 592-token prompt, first in a fresh process both times, peaked at 23.954 GB once and at 24.112 GB once, 0.16 GB apart [K, EV:RESULT_card_w24; EV:w24runs; BM:w24g]. Near 450-520 tokens the label is no guarantee [G].
  - The prompt pass sets the peak. Its MLX prefill memory grows with the prompt (median 22.76 GB up to 481 tokens, 23.95 GB above 1,024), and the lifetime footprint runs about 1.0 GB above it [K, EV:RESULT_card_w24].
  - On six long gate prompts (726-2,801 tokens), only 256-token chunks kept it under the line, and neither smaller chunk is the gated configuration: 512-token chunks still peaked at 24.345 GB, at 1.3-1.6x the first token, and 256-token chunks at 23.954 GB, at 2.0-2.9x. Their tokens differ from the gated configuration's on 6 / 6 and 5 / 6 prompts, and their logits hashes on all 6 [K, BM:w24e].

Prompts of 10K-30K tokens were not run with smaller chunks.

- **32 GB on long prompts.** On a 10K- and a 30K-token prompt the 32 GB chat configuration peaks at 34.0 and 35.5 GB [K, LP:data/prefill_1109.md]. The 12 GB line was not run on long prompts.

## 6.4 First token

- **At 12 GB the prompt pass streams most of the experts once.**
  - A 63-token prompt reads 21,986 MB in 2.906 s (21.3 tok/s) [K, F12H: chw18s, ifbench/10].
  - Code prompts of 103-367 tokens take 3.42-5.15 s [K, F12R:ex16].
- **Check 0** adds 0.3-0.8 s on average: the first draft plus a full verify [K, F12R: mean first token minus mean prefill, per run].
- **So the first token is 2.9 s on chat and 4.3 s on code** (medians). On code and chat, Strata's is 1.4-1.7 s at 24 GB and 2.0-2.8 s at its smallest settings [K, FG:low]. This is the largest loss at small memory.
- **At 24 GB (w13b)** the first token is 3.54 s on 12 HumanEval, 3.34 s on 12 MBPP, 3.09 s on chat and 2.86 s on math (medians) [K, BM:w24d; BM:w24f; BM:w24g], against Strata's 1.52-1.66, 1.31, 1.46-1.65 and 2.13-2.31 s on the same prompts at 24 GB [K, SE:h2h/results_low24]. On six long gate prompts (726-2,801 tokens) it is 5.7-14.6 s [K, BM:w24e]. On longer prompts it is 18.0 s at 3,003 tokens, 42.1 s at 6,980, 60.4 s at 9,988 and 195.0 s at 29,988 with the default prefill. The layer-major prefill ([MAIN] section 6.2) cuts that 2.4-2.7x (17.7 s at 6,980 tokens, 23.7 s at 9,988, 71.7 s at 29,988), for 0.3-2.4 GB more peak (26.54 GB at 10K, 30.60 GB at 30K) [K, LP:data/tue_afternoon_1110.md]. Its 256-token answer to the 6,980-token prompt (last chunk 836 tokens) is the default prefill's, character for character [K, same file]. It runs only on prompts over 1,024 tokens, and at 32 GB it is exact except when the last 1,024-token chunk is under about 200 tokens [K, LP:data/pfa_tail_1111.md]; the 24 GB line was not checked on such a prompt. Package 1.1.0's terminal chat, chat.py, runs it by default at 24 and 32 GB ([RELEASE] section 4).

## 6.5 Strata at 24 GB and at its smallest setting

Strata (MIT; github.com/Niko1221/Strata), version 0.1.58-metal, runs on the same machine and the same prompts in our harness [K, FG:low and its reports].

- Its weights are other files: 3-bit IQ3_XXS.
- Arm B reads the experts lossily. Arm D reads them exactly, at BF16, which Strata calls exact (its own label; not exact to our reference).
- Whole request is pooled tokens over request wall time, as in the note. Strata's reports also print an "END TO END" column. It is the geometric mean over requests of generated tokens / request wall time [SE:h2h/report.py, l. 8], e.g. 9.70 on chat at the smallest setting against the pooled 11.22.

| Strata setting | peak GB | set | whole decode | whole request | first token s | source |
|---|---|---|---|---|---|---|
| arena 16 GB, B (lossy) | 23.75 | code (12 HE + 12 MBPP) / chat / math | 26.4 / 28.4 / 28.0 | 21.4 / 21.0 / 23.0 | 1.41 / 1.65 / 2.31 | FG:low |
| arena 16 GB, D (Strata's exact) | 24.14 | code / chat / math | 26.6 / 26.9 / 26.7 | 21.0 / 20.3 / 22.1 | 1.40 / 1.46 / 2.13 | FG:low |
| arena 4 GB, B | 14.82 | code / chat / math | 15.47 / 15.47 / 15.41 | 12.89 / 11.93 / 13.19 | 1.99 / 2.68 / 3.23 | FG:low |
| arena 3.37 GB (smallest), B | 14.19 | chat | 14.42 | 11.22 | 2.78 | FG:low |

- **At 24 GB, on the same prompts** [K, BM:w24d; BM:w24f; BM:w24g; SE:h2h/results_low24]. Ours (the gated lossy configuration) against Strata's two arms, as whole decode, whole request and first token:
  - 12 HumanEval: 130.3 against 26.1-26.2 tok/s, 40.8 against 21.3-22.0, 3.54 against 1.52-1.66 s;
  - 12 MBPP: 91.2 against 27.0-27.3, 26.1 against 20.3-20.4, 3.34 against 1.31 s;
  - 12 IFEval (chat): 30.6 against 26.9-28.4, 18.1 against 20.3-21.0, 3.09 against 1.46-1.65 s;
  - 12 math: 40.8 against 26.7-28.0, 27.4 against 22.1-23.0, 2.86 against 2.13-2.31 s.
  - Peaks: ours 23.66-24.01 GB, Strata's 23.75-24.14 GB.
  - On code ours leads on decode (3.3-5.0x) and on whole request (1.3-1.9x), and on math by 1.5x and 1.2x. On chat Strata leads on whole request (1.1-1.2x), and our decode leads only 1.1x. Strata's first token is faster on every set: 1.9-2.5x on code and chat, 1.2-1.3x on math. HumanEval flatters our drafter (section 6.1).

- **Our exact 24 GB configuration** (E24: 41.1 HumanEval, 37.4 MBPP, pooled) leads Strata's code decode (26.4-26.6) by 1.4-1.6x [K]. This is rough: a different runner, per-set numbers, and E24's HumanEval did not stop at EOS. E24 has no whole request or first token in our standard metrics.
- **At 12 GB** Strata does not fit; its smallest setting peaks at 14.19 GB.
  - At that setting, with 3.2 GB more than our 10.96 GB, its chat beats ours: decode 14.42 vs 12.76 (+13 %), whole request 11.22 vs 10.09 (+11 %), first token 2.78 vs 2.92 s.
  - On code at a 4 GB arena (14.82 GB) its whole request is level with ours (12.89 vs 13.12) and its first token is better (1.99 vs 4.29 s). Our code decode leads (18.34 vs 15.47).
- **Why Strata wins on chat** is in section 11: it drafts one token per pass and verifies two rows.

# 7. Where a 12 GB verify call spends its time (measured)

The loop runs P2' + PR1 + the tail rule on 12 IFEval at cap 128, in two runs of the same 629 exact calls; the outputs are bit-identical [K, F12P].

- **Instrumentation.** Each call is stamped at four points: after the graph is built, after evaluation, after the host compare and after commit. The loader's own clocks give its busy, read and idle time; the main thread's CPU clock is read at the same points.
- **Fit.** Each stage is fitted as ms = a[w] + b x GB read, with w = rows per call.

| per exact call | 1 row | 2 rows | 3 rows | 4 rows | per GB read |
|---|---|---|---|---|---|
| whole call | 42.4 | 43.9 | 52.5 | 59.4 | 83.7 |
| loader reads | 8.9 | 9.7 | 13.8 | 15.6 | 81.4 |
| loader idle (nl) | 31.6 | 31.0 | 36.1 | 40.4 | 2.4 |
| main-thread CPU while the graph is built | 25.3 | 21.8 | 20.2 | 20.9 | 1.9 |
| nl minus that CPU (at least GPU side + handoffs) | 6.3 | 9.2 | 15.9 | 19.5 | 0.5 |

- **Reads are 64 % of chat time**, at about 12 GB/s. Each call also pays 9-16 ms of read latency on stalled layers.
- **The loader-idle time is 27 %:** 31-40 ms per call. Within it:
  - Python (main-thread CPU) is at most 21-25 ms. This is flat in width, about 0.45 ms per layer, and only partly on the critical path.
  - The GPU side and the handoffs are at least 6-20 ms, growing with width.
- **The loader's own overhead is 0.1 ms.** Commit is 1-2 ms, and drafts are 10.5 ms per check (6 %).
- **Repeatability:** the second run's whole-call fit matches the first's to 1 ms (43.3 / 43.6 / 53.2 / 58.8 + 82.7 per GB).

# 8. Corrected headlines and negative results

- **Withdrawn: "12 GB, 40.8 tok/s 'exact'" was a FAST arm**.
  - The 4 Nov results-first table listed it as "bitwise = its reference, 11.35 GB lifetime, 40.8 tok/s, 16 code-heavy prompts" [RF]. The line's owner corrected it on 6 Nov [F12 'CORRECTION for the shared plan']:
    - It was the full-output number of a two-pass arm. That arm commits rows certified by a cheap top-2 pass (theta 0.8) without the exact check.
    - Its tokens equal the verified loop's on 5 / 16 prompts.
  - The table's "likely first at <= 12 GB, exact" falls with it. The corrected 12 GB headline is the verified loop of table 6.1: 18.34 tok/s whole decode, 13.12 whole request, 4.29 s first token, 11.14 GB on 12 HumanEval + 12 MBPP.
  - No 12 GB configuration is exact.
- **The 12 GB FAST mode passed the map and failed the benchmark.**
  - It was priced at 45.8 tok/s [P] and passed the decision map's code line. Every theta tried then failed HumanEval+ (section 2).
  - The map's code line did not predict the benchmark; section 4 gives the map's side.
- **Negative results at 12 GB** [K, TR; each line carries its run]:
  - **Prefetch of predicted experts into victim slots** (October, the MTP path, pools of 51): -3 to +2 %.
    - 44 % of prefetches were unused, and the victims evicted experts the pass needed (+12 % waited).

    - A staging ring removed the victims but not the waste.
  - **A budget of 16 waited misses per pass:** +32 % code / +54 % prose, but it failed 3 of 5 benchmark sets against the 4-bit reference (IFEval -7.0). Lossy.
  - **LFU instead of LRU eviction:** +0.5 / -0.8 %.
  - **The read path is at the SSD floor.**
    - Reading base and sidecar rows in one burst: +0.5 %.
    - Direct reads into pool slots: -22 % (unaligned destinations).
    - Splitting each expert read into 2-6 reads: 1 aligned read per part takes the least time, and the loader is within ~5 % of the SSD burst floor.
  - **Computing resident experts before the SSD wait** (out-of-order experts): +0.6-1.3 %. Only 23 % of verify pairs were resident.
  - **Fewer or narrower drafts on the old MTP path:**
    - one draft per pass (k=1): -15 %
    - two drafts (k=2): +1.5 % code / +10 % prose, with changed tokens
    - a confidence threshold on the drafts: +0.8..+7.2 %
    - adaptive top-p per verify row: +4.3 % (noisy)
    - prompt lookup: +1.2 / -0.1 %
  - **GPU-side fusions, each under 1 %:** the one-dispatch recurrence, fused router ops, fewer command-buffer boundaries, multi-row projections (already at ~90 % of their bytes floor).
  - **Smaller expert stores** (all far above the error line, or small):
    - entropy coding of the streamed 3-bit part: 14.3 % at the bound, net <= +6 %
    - rotated re-quantization: 1.26x the 3-bit error at 3.25 bits
    - 2-bit codes (RTN / GPTQ / KLT): 3.65-7.7x the 3-bit error
    - a data-aware 2-bit trellis: 1.85-2.49x out of sample
    - a shared base expert: cosine to the layer mean ~0

# 9. Related work

- **Strata** (MIT; github.com/Niko1221/Strata), read in its source and docs [ST], is the engine we compared against.
  - It streams experts into an LRU arena with one F_NOCACHE read per expert from a contiguous copy of the experts, and keeps several reads in flight on I/O threads.
  - It reads ahead the experts that layer L+1's router predicts from layer L's input.
  - It drafts with the model's MTP layer, or with DFlash2 for some weight files. On streaming Macs of 32-48 GB, its setup turns the draft layer off.
  - We took nothing from its code. Our comparison and the levers it suggests are in sections 6.5 and 11.
- **Expert deferral.** Our one-token path's late-rank rule (section 3.2) is KTransformers' Expert Deferral (Chen et al., SOSP '25).
- **Weights streamed from flash.** Alizadeh et al., "LLM in a flash", run models larger than memory by reading weights from flash on demand.
- **Predicting and prefetching experts** for MoE inference with offloaded experts: Pre-gated MoE, MoE-Infinity, ProMoE, AdapMoE and HOBBIT [RW].
  - Strata's next-layer read-ahead belongs to this family. Our October prefetch, which lost (section 8), did too.
- **Low-memory runs of this model class, self-reported:**
  - SlotStream: 1.48 tok/s at a 10.7 GB process target, 4-bit, which it reports as exact; a user's run on a 16 GB Mac mini M2 with base storage (its issue #5; self-reported)
  - Palm-Infra: a different model, Qwen3.5-122B-A10B, at 12.4 tok/s at 5.9 GiB peak RSS with a 1 GiB expert cache on a 48 GB Mac; 4-bit (W4), exactness not stated (its docs; self-reported)
  - We did not run either.
- **Block drafting.** This line's drafter is our fine-tune of DFlash2 (Inco AI; Apache-2.0).
  - The harness loads the checkpoint incoai/Qwen3.8-Flash-Next-DFlash2 (revision a22a174, 19 Oct 2026; a finetune of Qwen/Qwen3.8-Flash-Next), then our fine-tuned weights over it.
  - DFlash 2 is described in Inco AI's "DFlash 2: Keep Drafting Parallel" (Aug 2026). The method is Chen, Liang and Liu, "DFlash: Block Diffusion for Flash Speculative Decoding" (ICML 2026).
  - Strata ships its own tuned copy. [DRAFTER] covers the drafters.
- **MTPLX** by Youssof Altoukhi (Apache-2.0; github.com/youssofal/mtplx) is the model code our engine builds on (section 3).

# 10. Limits

- **One machine, one model.** One prompt pass per arm on 12-24 prompts per set; no confidence intervals on speed.
- **Rough rows.** The share-lock rows ran beside other jobs (marked "rough"). The quiet rows are the headline.
- **The 12 GB lossy modes are not certified.** The two-pass mode failed; the width rule passed an IFBench-only test on loose scoring, and its strict score is significantly lower.
- **24 GB:** the gated configuration's speed was measured on 12 prompts each of HumanEval, MBPP, IFEval and math. Its memory line held in our runs on prompts of up to about 500 tokens, with little room (section 6.3). The exact E24 has no chat or math, and no whole request or first token in our standard metrics; its HumanEval decoded past EOS, and its whole decode is its report's per-pass figure, which our independent re-derivation did not cover.
- **32 GB:** chat only. The exact run's exactness is shown on tokens (12 / 12), not on logits. The lossy configuration is not gated at 32 GB: its gate did not fit the release day's machine time.
- **Missing at 12 GB:** math. The brief forbade new runs for the paper.
- **The memory line** is the engine's own footprint on a 64 GB machine, not an OS limit. On a real 12 GB machine, the OS and other apps also need room [G]. It holds on our speed sets except one 520-token math prompt at 24 GB (24.01 GB); longer prompts go over it at 24 and 32 GB, and the 12 GB line was not run on them (section 6.3).
- **SSD heat.** A small cache reads the SSD all the time. Strata's exact arm at a 4 GB arena stopped 7 times on SSD heat (>= 70 C) [K, FG:low]. Our 12 GB runs were not instrumented for SSD temperature [G].

# 11. What we left on the table (measured room)

- **Two-row verify calls on chat (C2c)** [P, F12C].
  - One draft per check, then 2-row calls until the first miss: what Strata does on chat with its one-token MTP, but keeping the block's later drafts.
  - A CPU replay of the measured calls prices it at 15.6-16.4 tok/s whole decode on chat, x1.10-1.15 over P2'. The replay reproduces the measured policies within 1.5 %.
  - On code it prices x1.01-1.02.
  - A 2-row call plus its draft costs 104 ms [K], already less than Strata's 115.7 ms pass. Screened after the release.
- **Next-layer read-ahead** (Strata's): predict layer L+1's experts from layer L's input and read them during layer L.
  - Upper bound: the loader-idle time (31-40 ms per call, 27 % of chat time) and the 9-16 ms read latency per call could hide behind reads [P, F12P].
  - Our own October prefetch (another predictor, small pools) wasted 44 % of its reads and lost. Strata's predictor and arena differ.
  - An engine change; not tried.
- **Python on the critical path:** at most 21-25 ms of main-thread CPU per call. Halving it is worth at most x1.09 [P, F12P]; a causal test was designed and not run.
- **The prompt pass reads 21,986 MB in 2.906 s for a 63-token prompt**, an effective 7.6 GB/s [K, F12H]. At the ~12 GB/s our verify calls reach, it would take about 1.8 s [P].
- **A chat drafter.** Chat keeps 2.3-2.4 tokens per check against code's 8.9-10.1 [K]. Companion paper [DRAFTER] covers the chat drafter work.
- **A lossy 24 GB chat route was never run on an engine.** It prices at 81.7 MB per kept token on IFEval, 61 % under the exact rule's 210.2 [P, 5.3]. It passes the map at 147 slots [K, 4.4], and would need its own gate.
- **A 24 GB prompt pass that stays under the line without changing the tokens.** 256-token chunks stay under but change them, and 512-token chunks go over (section 6.3).
- **Eviction.** The offline optimum (OPT) reads 38-55 % fewer exact bytes than LRU; at 12 GB, 386.8 vs 621.6 MB per kept token [P, 5.3]. No engine can reach OPT, so the gap bounds what a better eviction or forecast rule could take.
- **A 12 GB lossy chat rule.**
  - At 51 slots, the cf 0 rules fail the map's chat lines (IFEval 8.80 %, multilingual 5.26 %) [K, 4.4].
  - QOM cf 0.5 at 51 slots (216.2 MB per kept token [P]) is built as a map screen and was not run before the release.

- **The 45 GB chat check** ([MAIN]). After the byte cuts, SSD service is 5.7-6.3 ms of a 48.5-48.8 ms cycle [K, 5.6]. The SSD-free ceiling of today's check is ~57 tok/s [P].

# Appendix B. Source keys

Paths are relative to the root of the public snapshot, github.com/kamenlevi/350-tps; commit hashes are those of each folder's own repository, and the snapshot holds the files as last committed.

| key | file |
|---|---|
| F12R | `Fit12/data/paper/small_ram_12gb_rows.txt` (+ .json): every 12 GB row, by `Fit12/scripts/paper_rows.py` from the run logs `Fit12/data/scr/<run>/<tag>.txt` and their per-prompt phase logs. Runs: ex16, c1u16, tab12e, tab12f, chat12, chw13 (chw13a), chw14 (chw14c2, chw14ha, chw14h2) |
| F12 | `Fit12/PLAN_12GB_90_fit12.md`, by section heading |
| F12P | `Fit12/data/scr/chw16/chw16_prof.txt` and `Fit12/data/scr/chw15/chw15_prof.txt` (reader `Fit12/scripts/chw15_read.py`) |
| F12H | `Fit12/data/scr/chw18` (chw18_eq.txt, chw18_hist.txt, chw18s.txt) |
| F12C | `Fit12/scripts/rows_price.py` (the width-policy replay; inputs chw13 / chw14) |
| TR | `FableGod/TRIED/fit12.md` (the 12 GB ledger); October detail in `Fit12/PLAN.md` |
| EV:RESULT_t80ck1u | `EvalSuite/docs/RESULT_t80ck1u_2026-11-07.md` |
| BM:WED | `BigMoE/docs/WED_TESTS.md`, sections "E24, the 24 GB EXACT number" and "E24m, E24's MBPP12 with the EOS stop" |
| BM:c32 | `BigMoE/data/chs/c32_report.txt` |
| BM:w24d | `BigMoE/data/w24/w24d_report.txt` and `w24d_verdict.txt` (W24d: arms w13e and w13f = w13b; w14c and w14d = the 14-row variant); run records `BigMoE/data/r57/w13eh_100_runs.json` and `w13fh_100_runs.json` |
| BM:w24env | `BigMoE/data/stack/card_w24d_env.txt` (the gated 24 GB configuration's environment) |
| BM:w24f | `BigMoE/data/w24/w24f_report.txt` and `w24f_verdict.txt` (W24f: MBPP and IFEval; arms bf1 and bf2 = w13b, tf1 and tf2 = w13t); run records `BigMoE/data/r57/bf1m_100_runs.json`, `bf2m_100_runs.json`, `bf1i_100_runs.json` and `bf2i_100_runs.json` |
| BM:w24g | `BigMoE/data/w24/w24g_report.txt` (W24g: 12 math prompts, two passes, arms bg1 and bg2); run records `BigMoE/data/r57/bg1a_100_runs.json` and `bg2a_100_runs.json` |
| BM:w24e | `BigMoE/data/w24/w24e_report.txt` (prompt-pass chunks of 1,024, 512 and 256 tokens on six long gate prompts, 726-2,801 tokens); run records `BigMoE/data/r57/pc1024g_100_runs.json`, `pc512g_100_runs.json` and `pc256g_100_runs.json` |
| EV:RESULT_card_w24 | `EvalSuite/docs/RESULT_card_w24_2026-11-10.md` (w13b: PASS, 6 of 6 lines against NAX6-N; per-process lifetime peaks) |
| EV:RESULT_card_ifb12 | `EvalSuite/docs/RESULT_card_ifb12_2026-11-10.md` (the 12 GB width rule P2' on IFBench against NAX6-N: loose PASS, strict lower; one order and a 10-key reverse check) |
| EV:w24runs | `EvalSuite/results/card/es_cdw24_100136_fwd1_runs.json` to `es_cdw24_100136_fwd20_runs.json` and `es_cdw24_100136_rev1_runs.json` to `es_cdw24_100136_rev20_runs.json` (the gate's 40 processes; per prompt, the lifetime peak is the larger of mem[1] and mem[-1]) |
| LP | `Leap/` (here `Leap/data/prefill_1109.md`: the 32 GB chat configuration on 10K- and 30K-token prompts; `Leap/data/tue_afternoon_1110.md`: the 24 GB configuration on 3K- to 30K-token prompts, with run records in `Leap/data/tue_afternoon_1110/`) |

| key | file |
|---|---|
| FG:low | FableGod/findings/2026-11-08_1900_strata-low-memory.md and its reports FableGod/findings/2026-11-08_1900_strata_report_low24.md, FableGod/findings/2026-11-08_1900_strata_report_low12.md and FableGod/findings/2026-11-08_1900_strata_report_fit12.md |
| RF | MoonShots/research/results_first/RESULTS_FIRST.md (4 Nov 2026) |
| RW | MoonShots/research/relwork/RELWORK_CHECK.md (the related work's sources, with the form we cite, as listed in [MAIN] Appendix I) |
| SE:*path* | StrataEval/<path> (the Strata head-to-head harness and records; h2h/report.py gives the reports' columns) |
| BM:e24 / BM:e24m | BigMoE/data/e24/e24a.txt, e24b.txt and BigMoE/data/e24m/e24ma.txt, e24mb.txt (per-prompt lines; pooled here) |
| NI | NewIdeas/ (scripts/dm_oea_masks.py; NUMBER_CHECK.md, an independent re-derivation of each headline, by section) |
| ST | Strata's source upstream: Strata (MIT; github.com/Niko1221/Strata), commit 146df3c (metal/src/engine/stream.h, docs/MAC.md; its LICENSE) |
| SB | Breakthrough/SCOREBOARD.md, by heading |
| BD | Breakthrough/ (the data/ and scripts/ paths given after the key) |
| DM | FableGod/decision_map/ (dm_common.py, dm_ablations.py; outputs in out/) |
| NIQ | NewIdeas/QUEUE.md, by dated entry |
| EV:RESULT_chat_m1b | EvalSuite/docs/RESULT_chat_m1b_2026-11-08.md |
| EV:PREREG_relax | EvalSuite/docs/PREREG_relax_2026-11-02.md (revision 3); results in EvalSuite/docs/RESULT_relax_2026-11-05.md |
| BM:chs2 | BigMoE/data/chs/chs2_report.txt (arms x2c0, x2c0b, x2qm1, x2qm2, x2qm2s, x2e); lifetime peaks from the run records BigMoE/data/r57/<arm>_256_runs.json (the largest mem[6]) |
| BM:chs5 | BigMoE/data/chs/chs5_report.txt (arms z0, z0b); run records BigMoE/data/r57/z0_256_runs.json, z0b_256_runs.json |
| BM:pfd | BigMoE/data/pfd/pfd_report_0009.txt (arm pdz4a, chat as shipped; snapshot 36ee120) and pfd_report_2355.txt (its control pdz0); run records BigMoE/data/r57/pdz4a_256_runs.json, pdz0_256_runs.json |
| EV:PREREG_chat_gate | EvalSuite/docs/PREREG_chat_gate_2026-11-08.md (amendments 2 and 3: the MGSM line and its hold) |
| EV:PREREG_card_gates | EvalSuite/docs/PREREG_card_gates_2026-11-08.md (arm m1b; the typing counts in 'Typing') |
| EV:RESULT_card_m1b | EvalSuite/docs/RESULT_card_m1b_2026-11-09.md (M1-B + COH: PASS, 6 of 6 lines against NAX6-N) |
| EV:RESULT_card_lcoh | EvalSuite/docs/RESULT_card_lcoh_2026-11-09.md (L + COH: PASS, 6 of 6 lines against NAX6-N) |
| BM:ttft | BigMoE/data/chs/chat_ttft.txt |
| BM:git | the engine repository's commit 275c592 (the CHS5 arm list) and the frozen trees BigMoE/data/r57/snap/frozen_chs4 and BigMoE/data/r57/snap/frozen_chs5 |
| FG:chatbound | FableGod/findings/2026-11-07_2355_chat-bound-by-low-row-check.md |
| FG:c5 | FableGod/push/MISSION_C5_BYTES10.md |

# Use of AI tools

The code, the experiments and the text of this paper were produced with substantial help from AI coding agents. The author directed the work and is responsible for its content. The numbers were re-derived from the raw run records named beside them, and the references were opened by hand or by script.

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Companion papers:


- [DRAFTER] Kamen Levi. Drafting for a 125B mixture-of-experts model that does not fit in RAM: training its own MTP head, one drafter per text type, and the levers that failed. 2026. A companion paper of this set.
- [EVAL] Kamen Levi. Certifying lossy decoding on the model's known benchmarks: same-scores gates for a 125B mixture-of-experts model on one 64 GB Mac. 2026. A companion paper of this set.
- [MAIN] Kamen Levi. What It Takes to Run a 125B LLM Fast on a 64 GB Mac, and How to Go Faster. 2026. A companion paper of this set.
- [RELEASE] Kamen Levi. Running a 125B mixture-of-experts model on a 64 GB Mac: the engine package. 2026. A companion paper of this set.